

Algebra 1: Polynomials

Advanced Level

Name: _____

Date: _____

Question 1: Solve for x in the equation $x^2 + 5x - 6 = 0$

- A) $x = -2, x = 3$
- B) $x = -6, x = 1$
- C) $x = -3, x = 2$
- D) $x = 2, x = -3$
- E) $x = 6, x = -1$

Question 2: What is the degree of the polynomial $3x^4 - 2x^3 + x^2 - 1$?

- A) 1
- B) 2
- C) 3
- D) 4
- E) 5

Question 3: Factor the polynomial $x^2 + 7x + 12$

- A) $(x + 3)(x + 4)$
- B) $(x + 2)(x + 5)$
- C) $(x + 1)(x + 6)$
- D) $(x - 3)(x - 4)$
- E) $(x - 2)(x - 5)$

Question 4: Simplify the expression $(2x^2 + 3x - 1) + (x^2 - 2x - 3)$

- A) $3x^2 + x - 4$
- B) $3x^2 - x - 2$
- C) $3x^2 + x - 2$
- D) $x^2 + 5x - 4$
- E) $x^2 - 5x - 4$

Question 5: What is the remainder when the polynomial $x^3 - 2x^2 + x - 1$ is divided by $(x - 1)$?

- A) 0
- B) 1
- C) 2
- D) 3
- E) 4

Question 6: Find the x -intercept of the polynomial $2x^2 + 5x - 3$

A) $x = -3, x =$

$\frac{3}{2}$

A) $x = -1, x =$

$\frac{3}{2}$

A) $x = -$

$\frac{3}{2}, x = 1$

A) $x =$

$\frac{3}{2}, x = -$

$\frac{3}{2}$

A) $x =$

$\frac{3}{2}, x = -1$

Question 7: Solve the inequality $x^2 - 4x - 3 > 0$

A) $x < -1$ or $x > 3$

B) $x > -1$ and $x < 3$

C) $x < -3$ or $x > 1$

D) $x > -3$ and $x < 1$

E) $x < 3$ or $x > -1$

Question 8: Find the equation of the axis of symmetry for the polynomial $ax^2 + bx + c$

A) $x =$

$-\frac{b}{2a}$

A) $x =$

$-\frac{2b}{a}$

A) $x =$

$\frac{b}{2a}$

A) $x =$

$-\frac{a}{2b}$

A) $x =$

$\frac{2ab}{c}$

Question 9: Prove that the polynomial $x^3 + 2x^2 - 7x - 12$ can be factored as $(x + 3)(x^2 - x - 4)$

Question 10: Find the roots of the polynomial $x^4 - 2x^3 - 13x^2 + 14x + 12$ using the Rational Root Theorem and synthetic division

Chef's Tips (Hints):

- To factor a polynomial, look for common factors and use the distributive property
- To find the roots of a polynomial, use the Rational Root Theorem and synthetic division
- To solve an inequality, first solve the corresponding equation and then use the properties of inequalities to determine the solution set